

HW7

STAT 651

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1 BDA3: 5.5

Suppose $\theta = (\theta_1, \dots, \theta_J)$ can be written as $p(\theta) = \int \prod_{j=1}^J p(\theta_j|\phi)p(\phi)d\phi$

Prove that the covariances $Cov(\theta_i, \theta_j)$ are all nonnegative.

Because they are independent, $Cov(\theta_i, \theta_j|\phi) = 0$

Because they are identically distributed, if we let $X = E(\theta_i|\phi)$, then $E(\theta_i|\phi) = E(\theta_j|\phi) = X$

Also, the covariance of anything with itself is just the variance of that thing.

Use the marginal covariance:

$$\begin{aligned} Cov(\theta_i, \theta_j) &= E[Cov(\theta_i, \theta_j|\phi)] + Cov(E[\theta_i|\phi], E[\theta_j|\phi]) \\ &= E[0] + Cov(X, X) \\ &= Var(X) \end{aligned}$$

Because Variances are always positive, this shows that $Cov(\theta_i, \theta_j)$ is always nonnegative.

2 Hoff: 8.3

Fit a hierarchical model, as given on page 132, full conditionals given in the book:

$p(y|\theta_j, \sigma^2) \sim N(\theta_j, \sigma^2)$ (data for a school, each school has its own mean, variance is same across all)

$p(\theta_j|y_{i,j}, \dots, y_{n_j,j}, \sigma^2) \sim N(\frac{\frac{n_j \bar{y}_j}{\sigma^2} + \frac{1}{\tau^2}}{\frac{n_j}{\sigma^2} + \frac{1}{\tau^2}}, [\frac{n_j}{\sigma^2} + \frac{1}{\tau^2}]^{-1})$ (distribution of the thetas across all the schools)

$\frac{1}{\sigma^2} \sim \text{Gamma}(\frac{1}{2}[\nu_0 + \sum n_j], \frac{1}{2}[\nu_0 \sigma_0^2 + \sum_{j=1}^J \sum_{i=1}^{n_j} (y_{i,j} - \theta_j)^2])$

$\frac{1}{\tau^2} \sim \text{Gamma}(\frac{\eta_0 + J}{2}, \frac{\eta_0 \tau_0^2 + \sum (\theta_j - \mu)^2}{2})$

$p(\mu|\theta_1, \dots, \theta_J, \tau^2) \sim \text{Normal}(\frac{\frac{m \bar{\theta}}{\tau^2} + \frac{\mu_0}{\gamma_0^2}}{\frac{m}{\tau^2} + \frac{1}{\gamma_0^2}}, [\frac{m}{\tau^2} + \frac{1}{\gamma_0^2}]^{-1})$

Priors: $\mu_0 = 7, \gamma_0^2 = 5, \tau_0^2 = 10, \eta_0 = 2, \sigma_0^2 = 15, \nu_0 = 2$

a)

Run a Gibbs sampler to approximate the posterior distributions of $\{\theta, \sigma^2, \mu, \tau\}$. Assess convergence of Markov chain, find ESS for $\{\sigma^2, \mu, \tau^2\}$. Run chain until ESS for all are over 1000.

Below, I show the trace plots for the first 2 thetas, as well as the variables of interest. After that I provide the effective sample sizes, showing they are all over 1000.

I started by trying to implement Gibbs, but there's some bug in there (I only noticed once I plotted the posterior means vs. the sample means and noticed no borrowing of strength). So I switched to STAN.

```
# suppressPackageStartupMessages(library("coda"))
# suppressPackageStartupMessages(library("dplyr"))
# suppressPackageStartupMessages(library("invgamma"))
# set.seed(1337)
# dt <- read.csv('schools651.csv', header = TRUE)
# dt <- dt[, -1] # Remove the rownames
# mu_0 <- 7
# gamma2_0 <- 5
# tau2_0 <- 10
# eta_0 <- 2
# sigma2_0 <- 15
# nu_0 <- 2
# N <- 8 # Number of schools
# # Get average time per school
# y_bar_vec <- dt %>% group_by(school) %>% summarise(mean = mean(times, na.rm = TRUE))
# y_bar_vec <- y_bar_vec$mean
# # Get total students per school
# n_vec <- dt %>% group_by(school) %>% summarise(count = n())
# n_vec <- n_vec$count

# fc_theta <- function(n_vec, y_bar_vec, sigma2, tau2) {
#   mean <- ((n_vec * y_bar_vec/sigma2) + 1/tau2)/(n_vec/sigma2 + 1/tau2)
#   var <- 1/(n_vec/sigma2 + 1/tau2)
#   rnorm(N, mean, sqrt(var))
# }

# fc_sigma2 <- function(nu_0, sigma2_0, n_vec, theta_vec, dt) {
#   sum_sum_sqr_resid <- 0
#   for (i in 1:N) {
#     sum_sum_sqr_resid <- sum_sum_sqr_resid + sum((dt$times[dt$school == i] - theta_vec[i])^2)
#   }
#   shape <- 1/2*(nu_0 + sum(n_vec))
#   rate <- 1/2*(nu_0*sigma2_0 + sum_sum_sqr_resid)
```

```

#   rinvgamma(1, shape = shape, rate = rate)
# }

# fc_mu <- function(N, theta_vec, mu_0, tau2, gamma2_0) {
#   mean <- ((N*mean(theta_vec)/tau2) + (mu_0/gamma2_0))/(N/tau2 + 1/gamma2_0)
#   var <- 1/(N/tau2 + 1/gamma2_0)
#   rnorm(1, mean, sqrt(var))
# }

# fc_tau2 <- function(eta_0, tau2_0, theta_vec, mu) {
#   shape <- (eta_0 + N)/2
#   rate <- (eta_0*tau2_0 + sum((theta_vec - mu)^2))/2
#   rinvgamma(1, shape = shape, rate = rate)
# }

# n_iter <- 2000
# state <- list(theta = rnorm(N, mu_0, sqrt(sigma2_0)), sigma2 = sigma2_0, mu = mu_0, tau2 = tau2_0)
# sims <- matrix(NA, ncol = N + 3, nrow = n_iter)
# colnames(sims) <- c(paste0("theta", 1:N), "sigma2", "mu", "tau2")
# burn_in <- 100

# for (i in 1:n_iter) {
#   state$theta <- fc_theta(n_vec, y_bar_vec, state$sigma2, state$tau2)
#   state$sigma2 <- fc_sigma2(mu_0, sigma2_0, n_vec, state$theta, dt)
#   state$mu <- fc_mu(N, state$theta, mu_0, state$tau2, gamma2_0)
#   state$tau2 <- fc_tau2(eta_0, tau2_0, state$theta, state$mu)

#   sims[i, 1:N] <- state$theta
#   sims[i, "sigma2"] <- state$sigma2
#   sims[i, "mu"] <- state$mu
#   sims[i, "tau2"] <- state$tau2
# }

# sims <- sims[-c(1:burn_in), ]

# plot(sims[, "theta1"], type = 'l')
# plot(sims[, "theta2"], type = 'l')
# plot(sims[, "sigma2"], type = 'l')
# plot(sims[, "mu"], type = 'l')
# plot(sims[, "tau2"], type = 'l')

# print(paste0("sigma2 Effective Sample Size: ", coda::effectiveSize(sims[, "sigma2"])))
# print(paste0("mu Effective Sample Size: ", coda::effectiveSize(sims[, "mu"])))
# print(paste0("tau2 Effective Sample Size: ", coda::effectiveSize(sims[, "tau2"])))

```

```

suppressPackageStartupMessages(library("rstan"))
suppressPackageStartupMessages(library("coda"))
suppressPackageStartupMessages(library("invgamma"))
suppressPackageStartupMessages(library("knitr"))
dt <- read.csv('schools651.csv', header = TRUE)
dt <- dt[, -1] # Remove the rownames
mu_0 <- 7
gamma2_0 <- 5
tau2_0 <- 10
eta_0 <- 2
sigma2_0 <- 15
nu_0 <- 2
N <- 8 # Number of schools
n <- length(dt$school)

data_stan <- list(N = N, n = n, y = dt$times, grp = dt$school, nu_0 = nu_0, sigma2_0 = sigma2_0)

params <- c("theta", "mu", "sigma2", "tau2")

n_cores <- parallel::detectCores() # Count how many cores are available
options(mc.cores = n_cores)

fit_stan <- stan(model_code = readLines("q2.stan"),
                 data = data_stan, pars = params,
                 iter = 5000, warmup = 500, thin = 5, chains = n_cores)

```

Warning in readLines("q2.stan"): incomplete final line found on 'q2.stan'

```

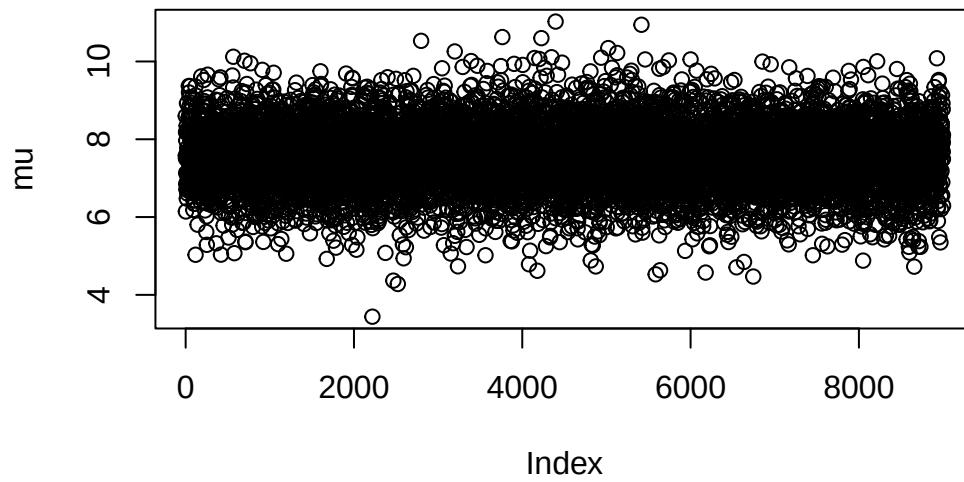
sim_stan <- extract(fit_stan) # collect samples from all chains

eff <- summary(fit_stan)$summary[, 9]

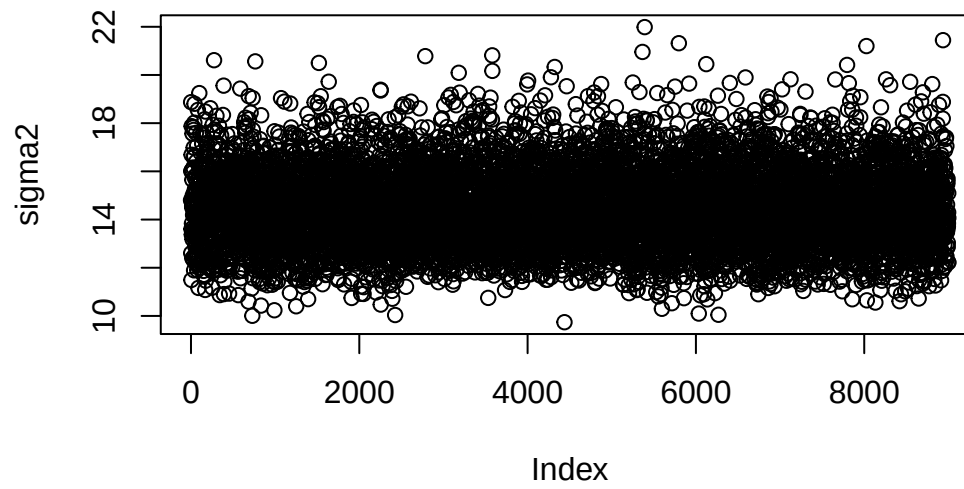
theta1 <- sim_stan$theta[, 1]
theta2 <- sim_stan$theta[, 2]
theta3 <- sim_stan$theta[, 3]
theta4 <- sim_stan$theta[, 4]
theta5 <- sim_stan$theta[, 5]
theta6 <- sim_stan$theta[, 6]
theta7 <- sim_stan$theta[, 7]
theta8 <- sim_stan$theta[, 8]
mu <- sim_stan$mu
sigma2 <- sim_stan$sigma2
tau2 <- sim_stan$tau2

plot(mu)

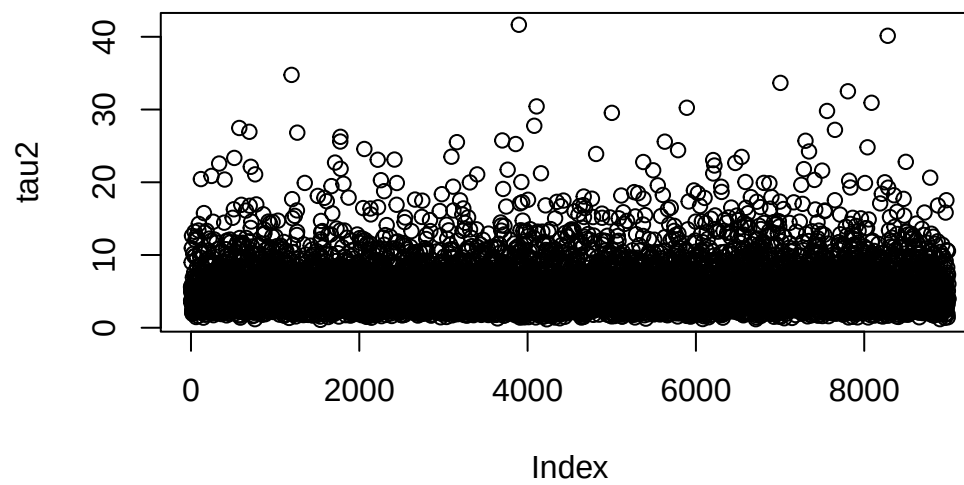
```



```
plot(sigma2)
```



```
plot(tau2)
```



```
kable(eff)
```

	x
theta[1]	8986.216
theta[2]	8649.214
theta3	8176.062
theta[4]	8922.080
theta[5]	9045.523
theta[6]	8813.018
theta[7]	9106.492
theta[8]	9054.884
mu	9055.791
sigma2	8585.959
tau2	8778.219
lp__	8143.285

The series appears to converge nicely (as shown in the trace plots), and quickly. I did a bunch of samples because STAN, so as shown the Effective Sample Size for all are above the target 1000.

b)

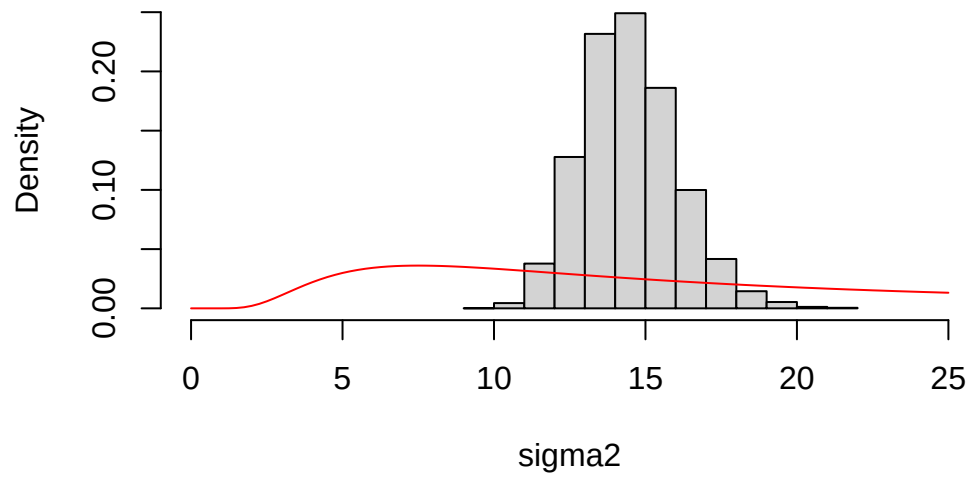
Compute posterior means and 95% confidence regions for $\{\sigma^2, \mu, \tau^2\}$. Also, compare the posterior densities to the prior densities, and discuss what was learned from the data.

```
post_mean_sigma2 <- round(mean(sigma2), 3)
post_ci_sigma2 <- round(quantile(sigma2, probs = c(0.025, 0.975)), 3)
post_mean_mu <- round(mean(mu), 3)
post_ci_mu <- round(quantile(mu, probs = c(0.025, 0.975)), 3)
post_mean_tau2 <- round(mean(tau2), 3)
post_ci_tau2 <- round(quantile(tau2, probs = c(0.025, 0.975)), 3)
```

Variable	Post. Mean	Lower	Upper
σ^2	14.486	11.736	17.852
μ	7.559	5.928	9.103
τ^2	5.43	1.944	14.003

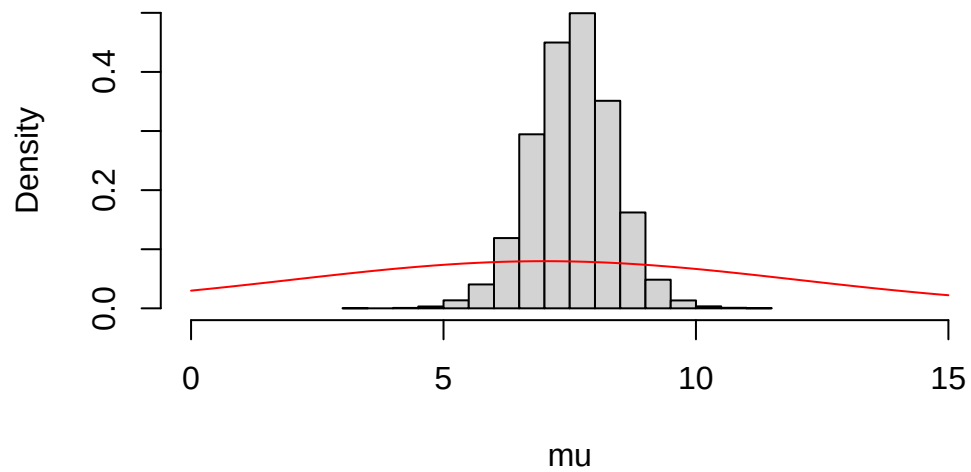
```
hist(sigma2, freq = FALSE, xlim = c(0, 25))
curve(dinvgamma(x, nu_0/2, rate = nu_0*sigma2_0/2), col = 'red', add = TRUE)
```

Histogram of sigma2



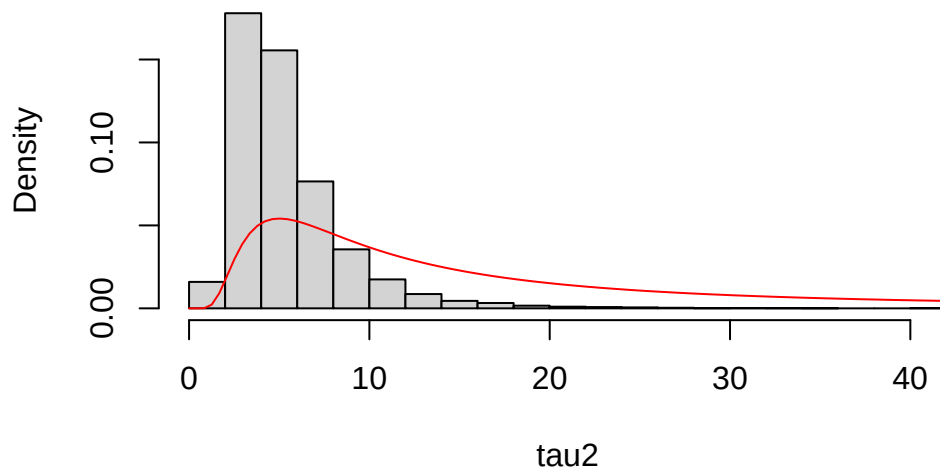
```
hist(mu, freq = FALSE, xlim = c(0, 15))  
curve(dnorm(x, mu_0, gamma2_0), col = 'red', add = TRUE)
```

Histogram of mu



```
hist(tau2, freq = FALSE)  
curve(dinvgamma(x, eta_0/2, rate = eta_0*tau2_0/2), col = 'red', add = TRUE)
```

Histogram of tau2



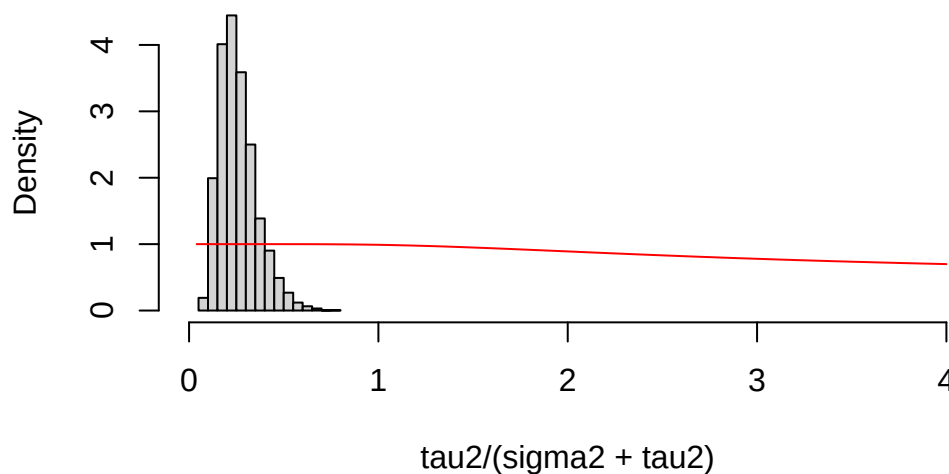
The posterior density for mu is much tighter than the prior (much lower variance). The opposite is true for tau2 where the posterior has a wider spread. The posterior for sigma2 has a completely shifted distribution compared to the prior but has a very long tail.

c)

Plot posterior density of $R = \frac{\tau^2}{\sigma^2 + \tau^2}$ and compare it to a plot of the prior density of R. Describe the evidence for between-school variation.

```
hist(tau2/(sigma2+tau2), freq = FALSE, xlim = c(0, 4))
curve(dinvgamma(x, eta_0/2, rate = eta_0*tau2_0/2)/(dinvgamma(x, nu_0/2, rate = nu_0*sigma2_0/2),
```

Histogram of tau2/(sigma2 + tau2)



The prior distribution for R is basically a uniform besides the blip at the start. The posterior shows a heavy concentration, meaning that there does appear to be a difference between the in-

school variance and the between-school variances (where as the prior assumes they are basically the same).

d)

Obtain the posterior probability that θ_7 is smaller than θ_6 , as well as the posterior probability that θ_7 is the smallest of all the θ 's.

```
less_than_6 <- round(mean(theta7 < theta6), 4)
smallest <- round(mean(theta7 < min(sim_stan$theta[, c(1:6, 8)])), 6)
```

Probability $\theta_7 < \theta_6 = 0.522$ Probability θ_7 is the smallest = 1.11×10^{-4}

e)

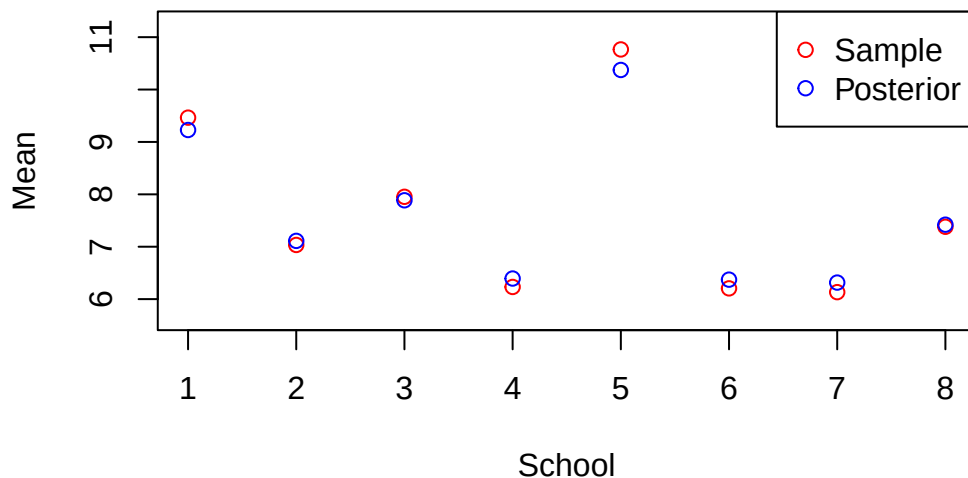
Plot the sample averages $\bar{y}_1, \dots, \bar{y}_8$ against the posterior expectations of $\theta_1, \dots, \theta_8$, and describe the relationship. Also compute the sample mean of all observations and compare it to the posterior mean of μ .

```
samp_means <- sapply(1:8, \ (x) mean(dt$times[dt$school == x]))
post_means <- sapply(1:8, \ (x) mean(sim_stan$theta[, x]))

overall_samp_mean <- round(mean(dt$times), 4)
mu_mean <- round(mean(mu), 4)

min <- min(min(samp_means), min(post_means))
max <- max(max(samp_means), max(post_means))
plot(1:8, samp_means, col = 'red', ylim = c(min - .5, max + .5), xlab = 'School', ylab = 'Mean')
points(1:8, post_means, col = 'blue')
legend(x = 'topright', legend = c('Sample', 'Posterior'), col = c('red', 'blue'), pch = 1)
```

Sample vs. Posterior Means



The posterior thetas are all more ‘interior’ (ie. closer to the population μ) than their sample values, indicating borrowing of strength due to a shared population mean.

The posterior mean is 7.5591, which is remarkably close to the sample mean of 7.6913. I suppose this makes sense since we assume they all come from a population mean, and only the schools should be what are significantly changing from the sample to posteriors.

3

Fit the change-point model for coal mining disasters reported in the slides. The data can be obtained either directly from the Carlin et al. (1992) paper, or processed from the boot R package, using code from the slides.

$$y_t | \theta, Z \sim \text{Pois}(\theta)$$

$$y_t | \lambda, Z \sim \text{Pois}(\lambda)$$

$$\theta | a_\theta, \gamma_\theta \sim \text{Gamma}(a_\theta, \text{rate} = \gamma_\theta)$$

$$\lambda | a_\lambda, \gamma_\lambda \sim \text{Gamma}(a_\lambda, \text{rate} = \gamma_\lambda)$$

$$Z \sim \text{Unif}(0, T)$$

$$\text{Use } a_\theta = a_\lambda = 0.5, \gamma_\theta \sim \text{Gamma}(2, 1), \gamma_\lambda \sim \text{Gamma}(1, 1)$$

$$L(\theta, \lambda, Z | y) = \prod_{t=1}^Z [\text{Pois}(y_i | \theta)] \prod_{t=z+1}^T [\text{Pois}(y_i | \lambda)]$$

$$f(\text{everything}) \propto \prod_{t=1}^Z [\text{Pois}(y_i | \theta)] \prod_{t=z+1}^T [\text{Pois}(y_i | \lambda)] \frac{\gamma_\theta^{a_\theta}}{\Gamma(a_\theta)} \theta^{a_\theta-1} e^{-\gamma_\theta \theta} \frac{\gamma_\lambda^{a_\lambda}}{\Gamma(a_\lambda)} \lambda^{a_\lambda-1} e^{-\gamma_\lambda \lambda} \frac{1}{T+1}$$

$$\begin{aligned} p(Z) &\propto L(\theta, \lambda, Z | y) \frac{1}{T+1} \\ &= \frac{1}{\prod_{t=1}^Z y_i!} \frac{1}{\prod_{t=z+1}^T y_i!} \theta^{\sum y_i} e^{-n_\theta \theta} \lambda^{\sum y_i} e^{-n_\lambda \lambda} \frac{1}{T+1} \end{aligned}$$

$$\begin{aligned}
p(\theta|everything) &= \prod_{t=1}^Z [Pois(y_i|\theta)] \frac{\gamma_\theta^{a_\theta}}{\Gamma(a_\theta)} \theta^{a_\theta-1} e^{-\gamma_\theta \theta} \\
&\propto \theta^{\sum y_i} e^{-n_\theta \theta} \theta^{a_\theta-1} e^{-\gamma_\theta \theta} \\
&\sim Gamma(\sum_{i=1}^Z y_i + a_\theta, n_\theta + \gamma_\theta) \\
\\
p(\lambda|everything) &= \prod_{t=z+1}^T [Pois(y_i|\lambda)] \frac{\gamma_\lambda^{a_\lambda}}{\Gamma(a_\lambda)} \lambda^{a_\lambda-1} e^{-\gamma_\lambda \lambda} \\
&\propto \lambda^{\sum y_i} e^{-n_\lambda \lambda} \lambda^{a_\lambda-1} e^{-\gamma_\lambda \lambda} \\
&\sim Gamma(\sum_{i=z+1}^T y_i + a_\lambda, n_\lambda + \gamma_\lambda)
\end{aligned}$$

a)

```

suppressPackageStartupMessages(library('boot'))
data("coal") # from the boot package on CRAN
yrs <- floor(coal$date)
year_seq <- min(yrs):max(yrs)
dat <- data.frame(year = year_seq,
                  count = as.numeric(table(factor(yrs, levels = year_seq))))
y <- dat$count

a_theta <- .5
a_lambda <- .5
T <- length(y)

theta_fc <- function(y, Z, gamma_theta) {
  shape <- sum(y[1:Z]) + a_theta
  rate <- Z + gamma_theta
  rgamma(1, shape = shape, rate = rate)
}

lambda_fc <- function(y, Z, gamma_lambda) {
  shape <- sum(y[(Z+1):T]) + a_lambda
  rate <- T-Z + gamma_lambda
  rgamma(1, shape = shape, rate = rate)
}

log_f_z <- function(y, Z, theta, lambda) {
  sum_theta <- sum(y[1:Z])
  n_theta <- Z
  sum_lambda <- sum(y[(Z+1):T])

```

```

n_lambda <- T-Z

fact_theta <- 1
fact_lambda <- 1
for (i in 1:length(y)){
  fact <- factorial(y[i])
  if (i <= Z) {
    fact_theta <- fact_theta * fact
  } else {
    fact_lambda <- fact_lambda * fact
  }
}

log(1/fact_theta * 1/fact_lambda * 1/(T+1) * theta^sum_theta * exp(-n_theta * theta) * lam
}

mh_z <- function(y, Z, theta, lambda) {
  # Process for getting a good candidate Z to explore full space
  prob_above <- (T-Z)/T
  above_or_not <- rbinom(1, 1, prob_above)
  if (above_or_not == 1) {
    prop <- rpois(1, (T-Z)/4)
  } else {
    prop <- Z - rpois(1, Z/4)
  }
  prop <- round(rnorm(1, Z, sd = 5))
  if (prop > 111 | prop < 2) {return(Z)}
  ratio <- log_f_z(y, prop, theta, lambda) - log_f_z(y, Z, theta, lambda)
  ifelse(log(runif(1)) < ratio, return(prop), return(Z))
}

n_iter <- 5000
state <- list(theta = .5, lambda = .5, gamma_theta = 1, gamma_lambda = 1, Z = 50)
sims <- matrix(NA, ncol = length(state), nrow = n_iter)
colnames(sims) <- c("theta", "lambda", "gamma_theta", "gamma_lambda", "Z")
burn_in <- 100

for (i in 1:n_iter) {
  state$gamma_theta <- rgamma(1, 2, 1)
  state$gamma_lambda <- rgamma(1, 1, 1)
  state$theta <- theta_fc(y, state$Z, state$gamma_theta)
  state$lambda <- lambda_fc(y, state$Z, state$gamma_lambda)
  state$Z <- mh_z(y, state$Z, state$theta, state$lambda)

  sims[i, 'theta'] <- state$theta
  sims[i, "lambda"] <- state$lambda
  sims[i, "gamma_theta"] <- state$gamma_theta

```

```

sims[i, "gamma_lambda"] <- state$gamma_lambda
sims[i, "Z"] <- state$Z
}

```

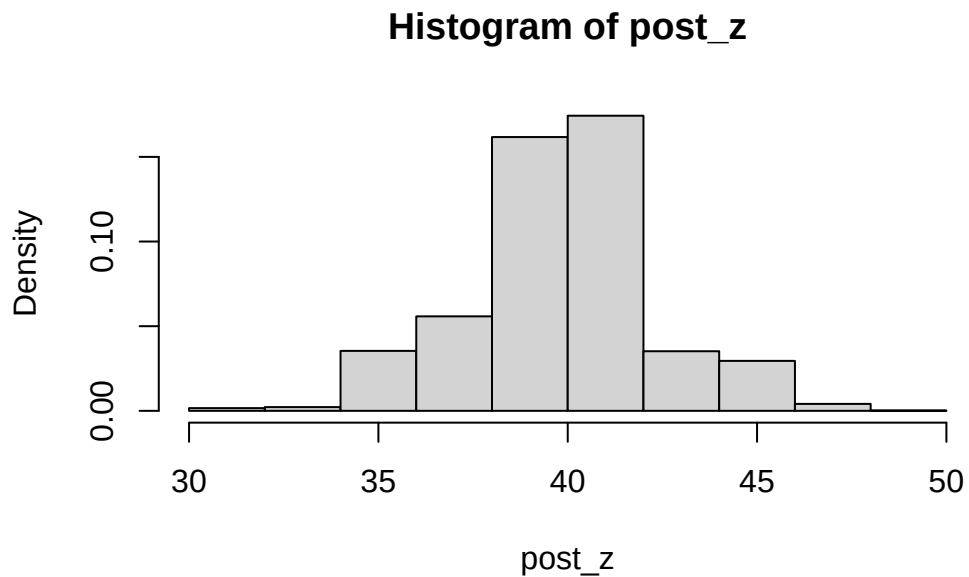
b

(i)

```

post_z <- sims[, "Z"]
hist(post_z, freq = FALSE)

```



```

u <- unique(post_z)
tab <- tabulate(match(post_z, u))
post_z_mode_ind <- u[tab == max(tab)]
change_year <- dat$year[post_z_mode_ind]

```

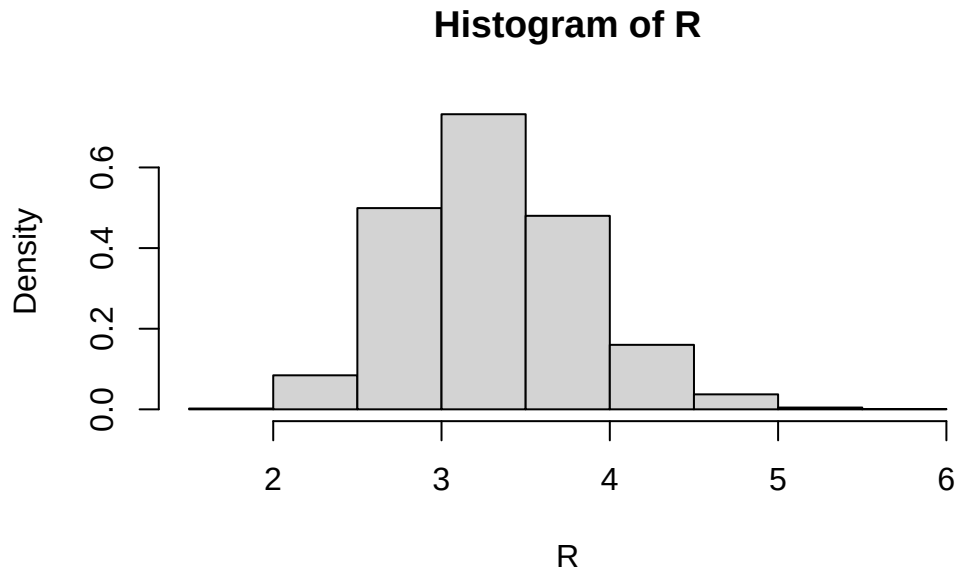
The posterior mode for Z is 41, which corresponds to a year of 1891.

(ii)

```

R <- sims[, "theta"] / sims[, "lambda"]
hist(R, freq = FALSE)

```



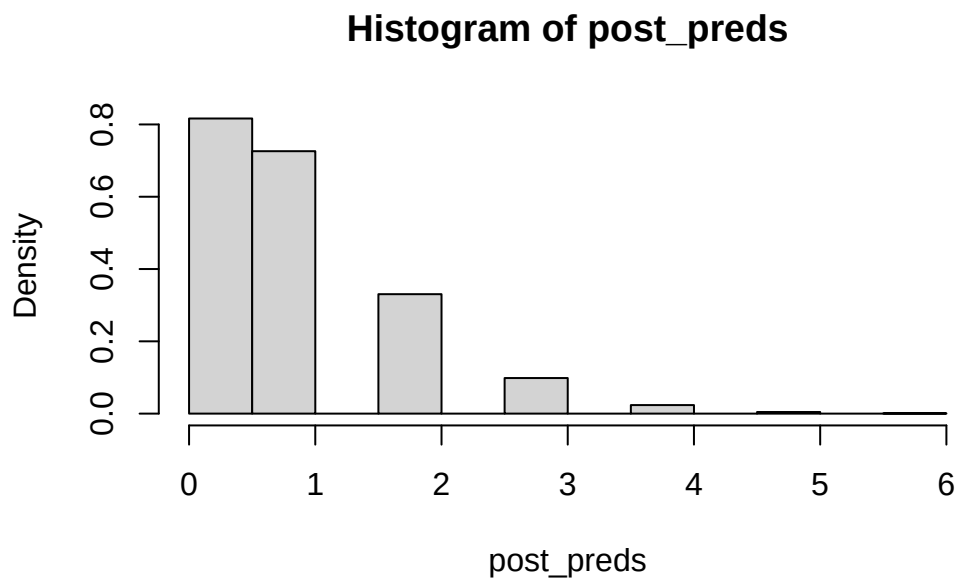
This plot represents the distribution of the ratio of the means prior to the change point and afterwards (ie. prior to the change point the mean was somewhere in the 3-4 range, and after it appears to be in the 0-1 range), giving ratios somewhere in the 3.5/1 ish range.

(iii)

```
# Lambda is parameter for second region
lams <- sims[, "lambda"]

# For all simulated lambda values, draw a corresponding prediction for a year
post_preds <- rpois(length(lams), lams)

hist(post_preds, freq = FALSE)
```



The histogram shows the posterior predictive distribution for 1963 (or really any year after the change point). Note that the first bar is the $y = 0$

Acknowledgements

Estimated time

I estimate this assignment took me in total around 10 hours to complete (mostly because I spent so long troubleshooting my Gibbs sampler on Q2 before switching to STAN).

Resources Used

I actually didn't really use any resources for this, besides just reminding me what the distribution PDFs are, and previous HWs to cannibalize the code.